HCF and LCM

Chapter 0 : Definition & Formulas

* LCM - least common Factor
  - It is the smallest positive number that is a multiple of two or more given numbers
  - Eg : LCM (4, 6)

    4 = 4, 8, 12, 16, 20, ...

    6 = 6, 12, 18, 24, ...

    ∴ LCM (4, 6) = 12

* HCF - Highest common factor
  - also known as Greatest Common Divisor (GCD)
  - It is the largest number that can divide two or more integers exactly, without leaving a remainder.
  - Eg : HCF (100, 125, 180)

    factors : 100 = 2 x 2 x 5 x 5

    125 = 5 x 5 x 5

    180 = 3 x 3 x 2 x 2 x 5

    ∴ HCF (100, 125, 180) = 5

Product of Two numbers = (HCF of the two numbers) x (LCM of the two numbers)

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$$\text{HCF of two numbers} = \frac{\text{Product of two numbers}}{\text{LCM of two numbers}}$$

$$\text{LCM of two numbers} = \frac{\text{Product of two numbers}}{\text{HCF of two numbers}}$$

Chapter 1 : How to solve quickly

LCM

2	80, 60
2	40, 30
5	20, 15
2	4, 3
2	2, 3
3	1, 3
	1, 1

Multiples of 80 = $2 \times \overset{2^4}{2 \times 2 \times 2} \times \underline{5}$

Multiples of 60 = $2 \times \underset{2^3}{2 \times 2} \times \underline{3} \times \underline{5}$

LCM = $2^4 \times 3 \times 5$

= 240 //

HCF

2	80	2	60
2	40	2	2
2	20	3	3
2	10	5	5
5	5		1
	1		

Multiple of 80 = $2 \times 2 \times 2 \times 2 \times 5$

Multiple of 60 = $2 \times 2 \times 3 \times 5$

HCF = $2 \times 2 \times 5$

= 20 //

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* HCF of the given numbers cannot be greater than any one of them
* LCM of the given numbers cannot be smaller than any of them
* If 1 is the HCF of 2 numbers,
  then their LCM will be their product
* For two co-prime numbers,
  their HCF is always 1
* mainly 4 types of questions :

Type 1 : Find the greatest / smallest number

1] The greatest possible length which can be used to measure exactly the lengths 5m, 4m, 12m 55 cm is

HCF (5, 4, 12m 55 cm)
= HCF (500, 400, 1255) cm

2	400
2	200
2	100
2	50
5	25
5	5
	1

5	500
5	100
5	20
2	4
2	2
	1

5	1255
251	251
	1

$400 = 2 \times 2 \times 2 \times 2 \times 5 \times 5$
$500 = 2 \times 2 \times 5 \times 5 \times 5$
$1255 = 251 \times 5$

∴ Highest possible length = 5

2] The HCF and LCM of two numbers are 10 and 560 respt., If one number is 70, find the other number.

Product of two numbers = HCF × LCM

$70 \times x = 10 \times 560$

$x = \frac{5600}{70} = 80$

∴ Other number = 80

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3] Find the greatest number which on dividing 1484 and 2045 leaves remainders 4 and 5 respt. ?

Required number = 1484 − 4
= 1480 //

Required number = 2045 − 5
= 2040

HCF (1480, 2040)

1480 = 2 × 2 × 2 × 5 × 37

2040 = 2 × 2 × 2 × 3 × 5 × 17

HCF = 2 × 2 × 2 × 5
= 40 //

Type 2 : Find HCF

1] Three numbers are in the ratio of 4 : 3 : 6 and their LCM are 3600. Find their HCF

let numbers be 4x, 3x and 6x

LCM = (4x × 3x × 6x)
= 12x //

4 = 2×2
3 = 3
6 = 3 × 2
LCM = 2×3×2 = 12

12x = 3600
x = 300 //

∴ numbers = 1200, 900, 1800

$1200 = 2^4 \times 3 \times 5^2$

$900 = 2^2 \times 3^2 \times 5^2$

$1800 = 2^3 \times 3^2 \times 5^2$

HCF = 2×2×3×5×5 = 300 //

2] Find the HCF of $\frac{2}{11}, \frac{4}{17}, \frac{6}{5}$.

$$HCF = \frac{\text{HCF of numerator}}{\text{LCM of Denominator}}$$

$2 = 2$
$4 = 2 \times 2$
$6 = 2 \times 3$
HCF $= 2$

$11 = 1 \times 11$
$17 = 1 \times 17$
$5 = 1 \times 5$
LCM $= 11 \times 17 \times 15$
$= 935$

$$HCF = \frac{2}{935}$$

Type 3 : Sum of two numbers is given, LCM & HCF is given, how to find reciprocals sum.

1] Sum of two numbers is 55 and the HCF and LCM of these numbers are 5 and 120 respt., then the sum of the reciprocals of the number is equal to :

$$a + b = 55$$

$$(a \times b) = HCF \times LCM$$

$$ab = 5 \times 120$$

$$ab = 600$$

Since, $\frac{1}{a} + \frac{1}{b} = \frac{a+b}{ab}$

$$\frac{1}{a} + \frac{1}{b} = \frac{55}{600} = \frac{11}{120}$$

$\therefore$ Sum of reciprocals $= \frac{11}{120}$

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Type 4 : Problem related to finding the biggest container to measure quantities

1] Suppose there are three different containers, containing different quantities of a mixture of milk and water whose measurement are 403 L, 403 L and 465 L. What biggest measure must be there to measure all the different quantities exactly?

HCF (403, 434, 465)

$403 = 13 \times 31$

$434 = 2 \times 7 \times 31$

$465 = 3 \times 5 \times 31$

HCF = 31 //

∴ biggest measure req. = 31 L //

Type 5 : Bell ring problems

1] Six bells commence tolling together and toll at intervals of 2, 4, 6, 8, 10 and 12 seconds respt. In 30 minutes, how many times do they toll together?

LCM (2, 4, 6, 8, 10, 12)

$2 = 1 \times 2$

$4 = 2 \times 2$

$6 = 3 \times 2$

$8 = 2 \times 2 \times 2$

$10 = 5 \times 2$

$12 = 3 \times 2 \times 2$

LCM $= 2 \times 2 \times 2 \times 3 \times 5$

$= 120$ //

Bells will toll together after every 120 s = 2 mins //

In 30 mins,

no. of times bell toll together

$= \frac{30}{2} + 1 = 16$ //

Type 6 : Circle based runner problem

1] Two people P and Q start running towards a circular track of length 400 m in opposite direction with initial speeds of 10 m/s and 40 m/s respt. Whenever they meet, P's speed double and Q's speed halves. After what time from the start will they meet for the third time?

Time taken to meet 1'st time

$= \frac{400}{10+40} = 8$ sec //

Now, P's speed = 20 m/s

Q's speed = 20 m/s

Time taken to meet 2nd time

$= \frac{400}{20+20} = 10$ sec //

Now, P's speed = 40 m/s

Q's speed = 10 m/s

Time taken to meet 3rd time

$= \frac{400}{10+40} = 8$ sec //

∴ Total time = 8 + 10 + 8

= 26 seconds //

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Type A : Greatest number

1] Find the greatest 5 digit number divisible by 5, 15, 20 and 25

lcm (5, 15, 20, 25)

$5 = 5 \times 1$ $\qquad$ $25 = 5 \times 5$

$15 = 5 \times 3$

$20 = 5 \times 2 \times 2$

$\text{lcm} = 5 \times 5 \times 3 \times 2 \times 2$

$= 300$

Greatest 5 digit number = 99999

$\frac{99999}{300} = 333.3$

So, Greatest required number

$= 333 \times 300$

$= 99900$

2] The product of two numbers is 3888. If the HCF of these numbers is 36, then the greatest number is.

let numbers be $36x, 36y$

$36x \times 36y = 3888$

$xy = \frac{3888}{36 \times 36} = 3$

Co-primes with product 3 = (1×3)

$\therefore$ Required no. $= 36 \times 1 = 36$

$= 36 \times 3 = 108$

$\Rightarrow$ Greatest no. = 108